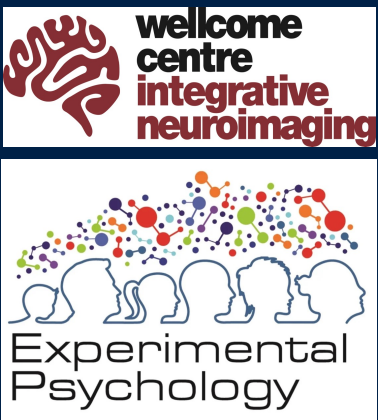


Neural activity associated with learning novel word forms in developmental language disorder



Nilgoun Bahar¹, Gabriel J. Cler², Saloni Krishnan^{3*} & Kate E. Watkins^{1*}

¹ Department of Experimental Psychology & Wellcome Centre for Integrative Neuroimaging, University of Oxford
² Department of Speech & Hearing Sciences, University of Washington
³ Department of Psychology, Royal Holloway, University of London
* Joint senior authors

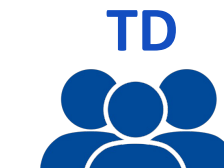


Background & Research Questions

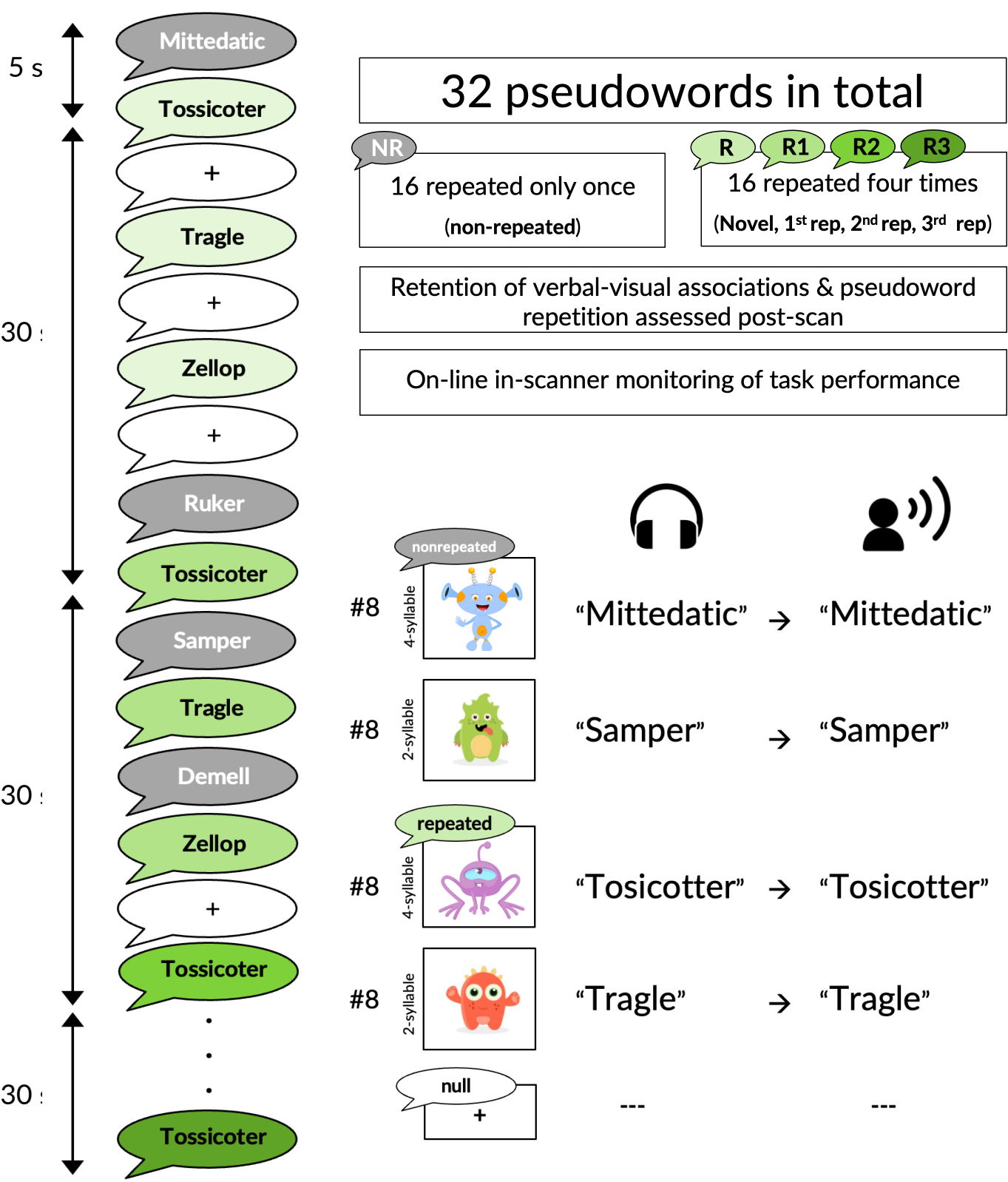
Children with **developmental language disorder (DLD)** perform poorly on tasks that require implicit learning of sequence structure, such as pseudoword repetition, which is a crucial skill for word learning & fluent speech. Learning pseudowords is associated with decrease in neural activity in left frontostriatal areas in adults, but the neural correlates of this skill are not well understood in typically developing (TD) children & those with DLD.

What are the neural correlates of novel word form learning in children? Are the brain activation patterns different in children with DLD relative to children with typical language development?

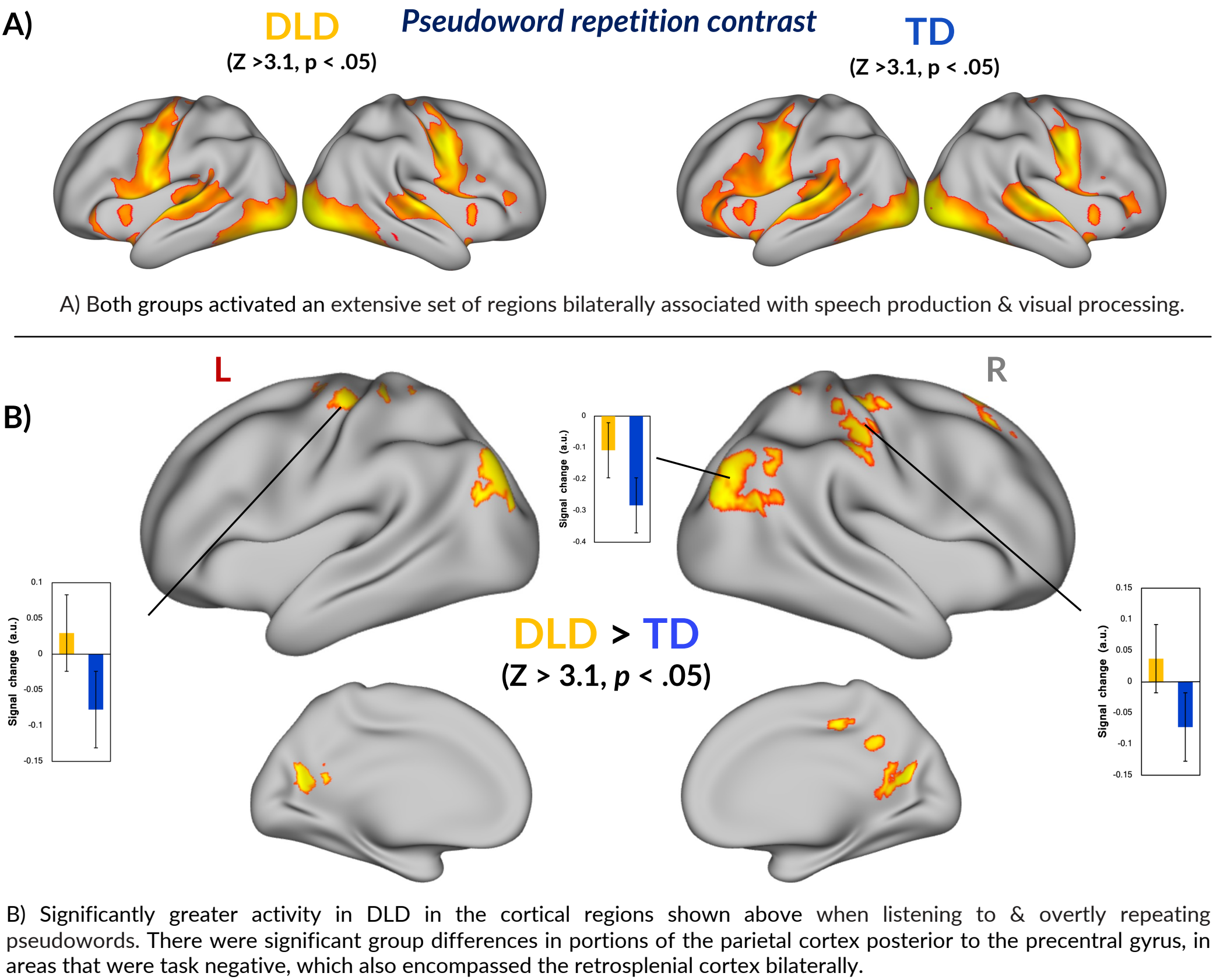
Participants & Group Definition

DLD		TD
 n = 46 m age = 12.36 y	118 native English-speaking, school-age children 10-16 years (post quality control)	 n = 72 m age = 12.52 y
	Nonverbal IQ > 70	
	1 ≥ SD below mean on 2+ language tests	
	History of speech & language difficulties	

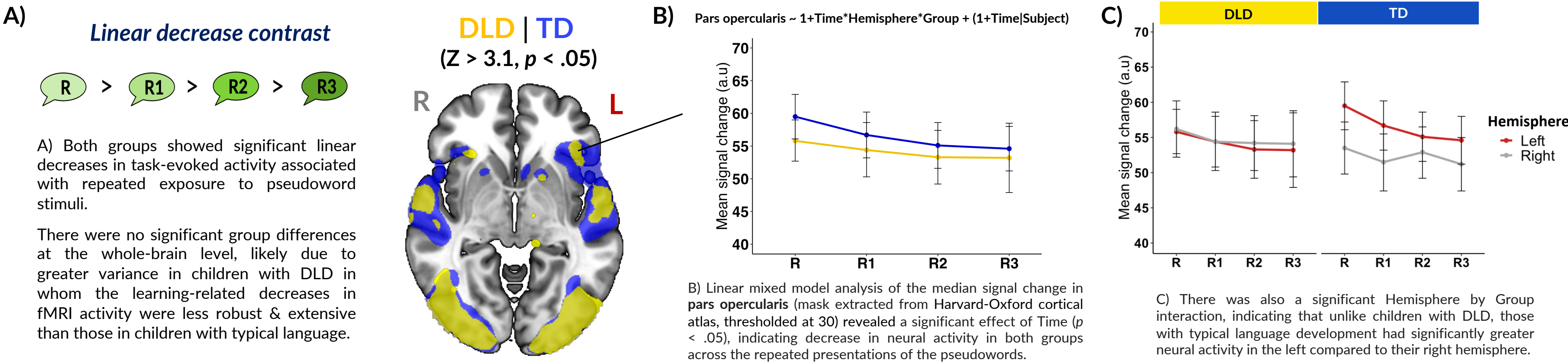
Study Design & Methods



fMRI Results: Articulating Novel Word Forms



fMRI & ROI Results: Learning Novel Word Forms



@NilgounBahar

nilgoun.bahar@psy.ox.ac.uk

<https://boldstudy.wordpress.com>

Summary & Take-home Messages

When producing novel word forms heard auditorily, children with DLD show significantly greater activity in the primary motor cortex bilaterally, at the level of hand area. This could reflect less finely tuned cortical specialization for speech articulatory control. Children with DLD also show less task-negative activity in parietal areas relative to typically developing children, possibly due to a less effective disengagement from the default mode network.

In both groups, learning novel word forms is associated with decreasing neural activity in the inferior frontal cortex & superior temporal cortex bilaterally, & subcortically in the dorsal striatum. These activity patterns resemble learning-related changes for novel word forms in adults but are more bilaterally distributed.